

Business Analytics – SAS Base

Lecture 4: Data Formats and Deal with dates

Aim of the lecture

- Show you different data formats in SAS and how to convert from one to another
- Demonstrate how to use data functions in SAS to process date variables efficiently

Agenda

- **Data format (informat vs. format)**
- Date functions
- Recap

Dealing with different SAS data formats **fluently** is a cool skill

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
1	ID	Num	8	BEST12.	BEST32.
8	age	Num	8	BEST12.	BEST32.
5	chas	Num	8	BEST12.	BEST32.
2	crim	Num	8	BEST12.	BEST32.
9	dis	Num	8	BEST12.	BEST32.
15	edu_ind	Char	4		
14	fancy_house	Char	1		
4	indus	Num	8	BEST12.	BEST32.
13	lstat	Num	8	BEST12.	BEST32.
6	nox	Num	8	BEST12.	BEST32.

Informats

Determines the data type of input data when reading the data into SAS

Formats

Determines the data type of output data or format data in reports

SAS informats can be grouped into three categories: character, numeric, and date/time

Informats	Syntax structure	Example
Character Informats	<i>\$Informatw.</i>	\$20.
Numeric Informats	<i>Informatw.d</i>	8.2
Date/Time Informats	<i>Informatw.</i>	mmddyy10.

Input function

INPUT function within a DATA step can be leveraged to convert the data type

```
data_newtype = input(data_oldtype, informats);
```

Formats determine how the output data looks like via FORMAT or PUT function

Formats

FORMAT variable data_format;

The detailed list of formats can be found in the following link:

<https://support.sas.com/documentation/cdl/en/leforinforref/64790/HTML/default/viewer.htm#p0z62k899n6a7wn1r5in6q5253v1.htm>

Put

data_newtype = put(data_oldtype, format)

You are not alone if as of now you find INPUT and PUT hard to differentiate

	INPUT	PUT
Target format	INPUT() can create different variables based on the informat	PUT() always create character variables
Source format	Source variable for INPUT() must always be character	Source format must match the source variable type in PUT()

Do not worry... let us try out some examples

Agenda

- Data format (informat vs. format)
- **Date functions**
- Recap

SAS has a LOT of different date formats

Informats for date variables	
Informats	Example
DATEw.	17MAR00
DATE9.	17MAR2000
DDMMYYw.	170300
DDMMYY8.	17032000
MMDDYYw.	031700
MMDDYY10.	03172000
YYMMDDw.	000317
YYMMDD10.	20000317

Formats for date variables	
Formats	Example
DATE7.	17MAR00
DATE9.	17MAR2000a
DDMMYYw.	17/03/00
DDMMYY10.	17/03/2000
MMDDYYw.	03/17/00
MMDDYY10.	03/17/2000
YYMMDDw.	00-03-17
WEEKDATEw.	Friday, March 17, 2000

Knowing the important ones is enough, you have Google any way...

Date functions are pretty intuitive (yeah, they do exist!)

Function	Syntax structure	Result
Year	YEAR(date)	Year
Month	MONTH(date)	Month
Day	DAY(date)	Day of month
Weekday	WEEKDAY(date)	Day of week
Quarter	QTR(date)	Quarter

INTNX is used to increment SAS date by a specified number of intervals

Method

INTNX(interval, start_date, increment, *alignment*)

Interval is the unit of measurement ('day', 'week', 'month', etc.)

Start_date is the starting point of the date

Increment is the number of intervals to be added

Alignment determines where the datevalue is aligned ('beginning' (default), 'middle', 'end', 'sameday')

INTCK calculates the difference between two dates

Method

INTCK(interval, start_date, end_date, *method*)

Interval is the unit of measurement ('day', 'week', 'month', etc.)

Start_date is the starting point of the date

End_date is the ending point of the date

Method determines the method to calculate the difference ('continuous', 'discrete')

Recap: Now you know how to deal with different SAS informats and formats

- Informats determine the data type when reading the data while formats determine the output data
- There are plenty of formats/informats for date variables and functions that can be leveraged to deal with date

*Thank
you*

